

FINAL PROJECT REPORT

SMARTPHONE USAGE, DIGITAL ADDICTION & WELL-BEING ANALYTICS

Executive Summary

This project develops an interactive Power BI analytics report for understanding smartphone usage behaviour and digital well-being. The uploaded PBIX file contains a smartphone_usage table and multiple report pages covering usage intensity, sleep, work/study time, gaming, social-media activity, notifications, stress, age group, gender and addiction status.

The report combines KPI cards, slicers, bar/column charts, pie and donut charts, a scatter plot, a decomposition tree and a Key Influencers visual. These components allow users to move from overall KPIs to demographic comparisons and then to possible behavioural drivers.

The analysis is exploratory. The visuals support comparison and pattern identification; they do not by themselves prove causation.

1. Introduction

1.1 Project Overview

Smartphones are used for communication, entertainment, education, work, social networking and many other activities. This project converts smartphone usage records into an interactive Power BI report for examining usage intensity, sleep, study/work time, stress and addiction.

1.2 Objectives

- Analyze daily and weekend screen time.
- Study usage behaviour across age groups and gender.
- Compare gaming, social-media usage, app openings and notifications.
- Examine screen time together with sleep and work/study hours.
- Analyze addiction levels and addicted/non-addicted status.
- Explore stress level and academic/work impact.
- Provide interactive filtering and analytical visuals.

2. Project Initialization and Planning Phase

2.1 Define Problem Statement

Raw smartphone usage records contain many behavioural variables, but individual columns alone do not provide an easy overall view. The project organizes these records into a dashboard that can answer practical questions about usage intensity, addiction, demographics, sleep, stress and daily activities.

2.2 Project Proposal (Proposed Solution)

The proposed solution is an interactive Power BI report with prepared fields, KPI cards, charts, slicers, a decomposition tree and Key Influencers analysis.

2.3 Initial Project Planning

Step	Activity	Purpose
1	Understand dataset	Identify fields, data types and analytical dimensions.
2	Data preparation	Check quality, categories and derived fields.
3	Model/report design	Create calculated categories and report pages.
4	Visualization	Build KPIs, charts, slicers and analytical visuals.
5	Validation	Check filters, aggregations and consistency.
6	Documentation	Prepare insights, conclusions and final report.

3. Data Collection and Preprocessing Phase

3.1 Data Collection Plan and Raw Data Sources Identified

The project is based on the smartphone_usage dataset embedded in the uploaded PBIX file. The Power BI report references the entity smartphone_usage smartphone_usage. The analysis uses user-level and behaviour-related attributes such as age, gender, screen time, sleep, work/study time, gaming, social media, notifications, stress and addiction.

3.2 Data Quality Report

The report uses categorical dimensions and numeric aggregations. A formal row-by-row missing-value audit is not exposed by the PBIX report definition, so this report does not invent missing-value percentages.

Area	Details
Categorical fields	gender, stress_level, addiction_level, addicted_label, academic_work_impact, Age Group, Addicted status
Numeric behaviour fields	age, daily_screen_time_hours, weekend_screen_time, sleep_hours,

	work_study_hours, app_opens_per_day, gaming_hours, social_media_hours, notifications_per_day
Identifier fields	user_id, transaction_id
Quality checks recommended	Missing values, duplicate IDs, invalid ranges, category spelling and outliers

3.3 Data Exploration and Preprocessing

Two derived categorical fields are visible in the report: Age Group and Addicted status. Numeric variables are aggregated with Count, Average and Sum depending on the visual.

Age Group = SWITCH(TRUE(), age<20, "Under 20", age<=29, "20-29", age<=39, "30-39", "40+")

Addicted Status = IF([addicted label]=1, "Addicted", "Not Addicted")

High Screen Time User = IF([daily screen time]>6, "High Usage", "Normal Usage") = 4960 users 66.13%

4. Data Visualization

4.1 Framing Business Questions

1. How much time do users spend on smartphones each day and on weekends?
2. How does screen time vary across age groups and gender?
3. How are users distributed across addiction levels and addicted status?
4. How do gaming and social-media activities compare?
5. How are screen time and sleep displayed together?
6. How does stress level relate to app-opening behaviour?
7. Which fields are highlighted as possible drivers in the Key Influencers analysis?
8. How can filters be used to investigate specific user segments?

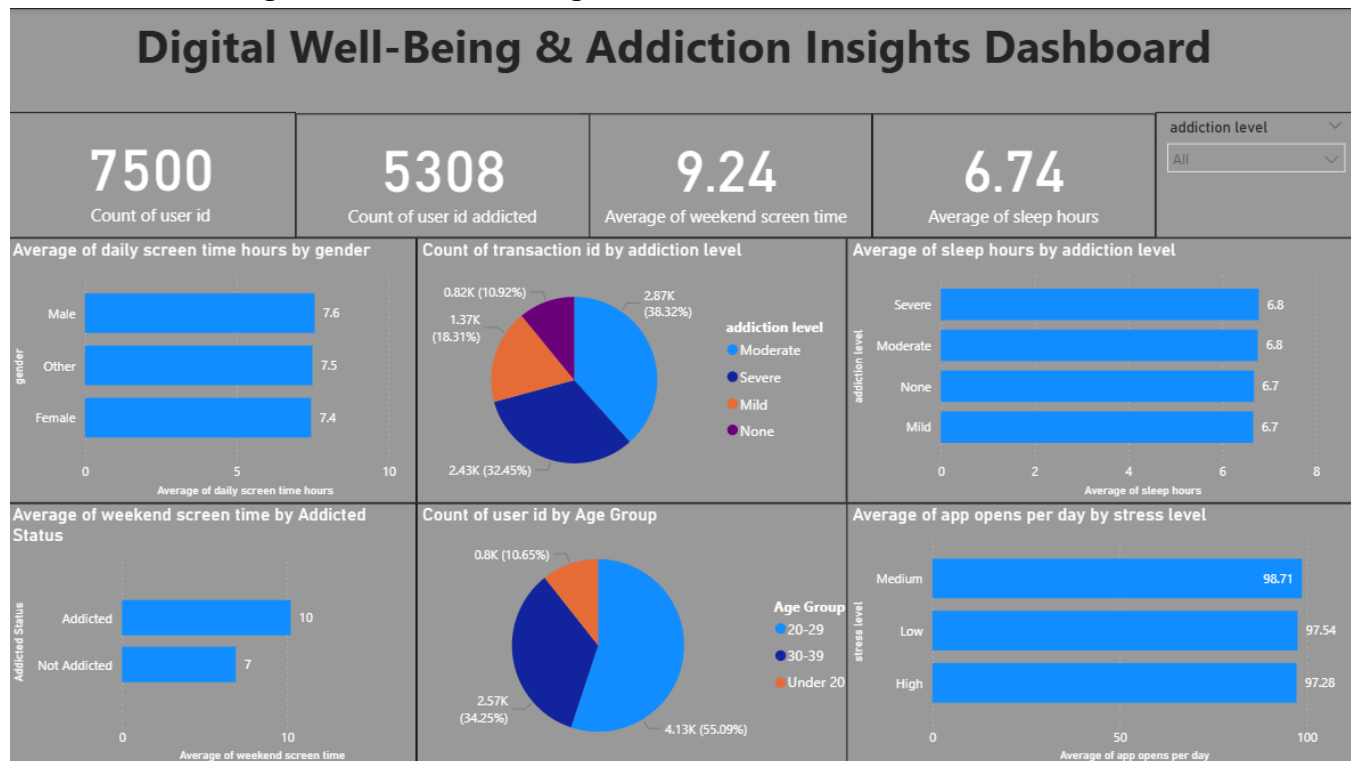
4.2 Developing Visualizations

Visual	Fields / Measures	Purpose
KPI cards	Daily screen time, work/study hours, user count, app opens, age, addicted label, weekend screen time, gaming, notifications, sleep, social media	Quick summary
Clustered bar/column charts	Addiction level, gender, age group, stress level and activity measures	Compare groups
Scatter chart	Daily screen time vs sleep hours	Inspect relationship/patterns
Pie/Donut charts	Age group and addiction/addicted status	Show distribution
Decomposition tree	User count by age group, gender, stress, addiction and academic/work impact	Drill into segments

Key Influencers	Usage and demographic fields against addicted status	Explore possible drivers
Slicers	Gender, stress level, addiction level, age group	Interactive filtering

5. Dashboard

5.1 Dashboard Design Screenshot with Insights



The uploaded PBIX contains a dedicated Dashboard page titled Digital well-Being & Addiction Insights Dashboard. It combines KPI cards, demographic charts, addiction distributions and interactive filtering.

Dashboard area	PBIX visual	Insight supported
Top KPI area	User count / sleep / weekend screen-time cards	Overall snapshot
Demographic section	Age-group and gender charts	Segment comparison
Behaviour section	Stress vs app opens; gender vs screen time	Behaviour comparison
Addiction section	Addiction-level distribution	Category distribution
Filter area	Addiction-level slicer	Focused analysis

Submission note: insert a rendered screenshot of the Dashboard page from Power BI Desktop here if your institution requires an actual screenshot.

6. Report

6.1 Report Design Screenshot with Insights

Report 1 focuses on demographic and addiction segmentation, including a decomposition tree. Report 2 focuses on digital well-being and possible drivers using Key Influencers. Key Influencers Result: When daily screen time increases by +2.61 hours, likelihood of addiction increases by 25.75x (25.75 times), and social media +1.59 hrs increases by 10.21x - from my Report 2.

Dashboard

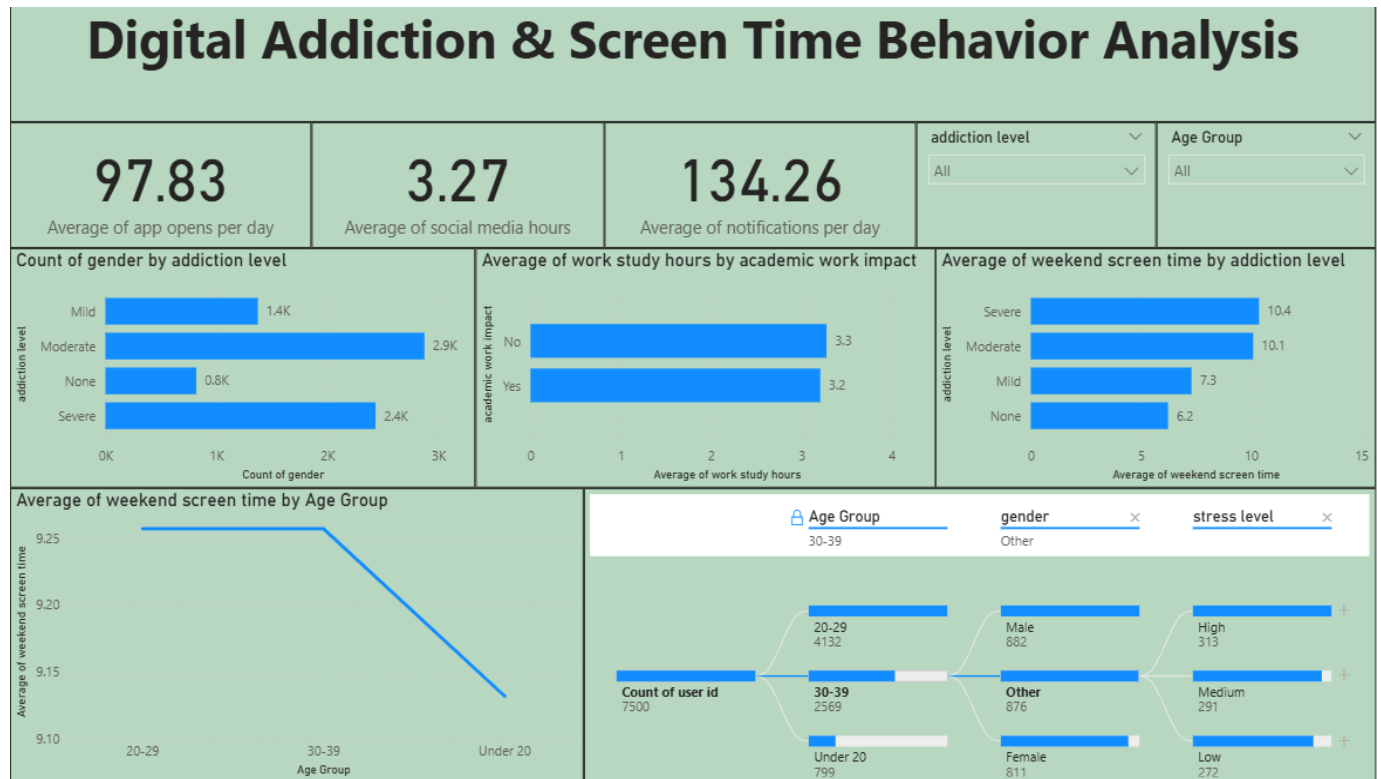
- User count, sleep and weekend screen-time KPI cards
- Age-group and addiction distributions
- Average screen time by gender
- Average app opens by stress level
- Addiction-level slicer

KPI cards

- Average daily screen time
- Average work/study hours
- User count
- Average app opens per day
- Average age
- Sum of addicted label
- Average weekend screen time
- Average gaming hours
- Average notifications per day
- Average sleep hours
- Average social-media hours

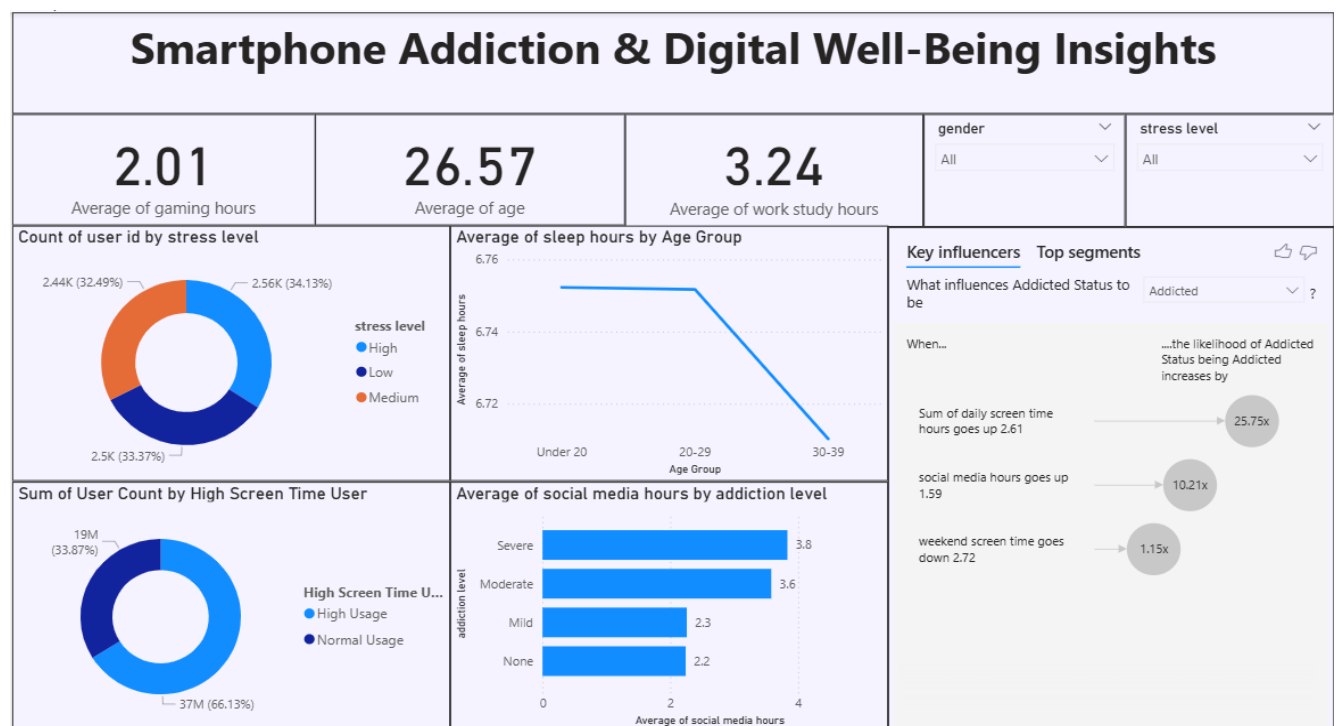
Report 1

- Addiction-level slicer
- Age-group slicer
- Gender distribution
- Charts for usage patterns
- Decomposition tree using age group, gender, stress level, addiction level and academic/work
- Impact



Report 2

- Gender slicer
- Stress-level slicer
- Key Influencers visual
- Charts for screen-time, social-media, gaming and weekend screen-time drivers
- Donut charts and KPI cards



vizualization

- Addiction-level distribution
- Daily screen time versus sleep hours scatter plot
- Gaming versus social-media comparison
- Average screen time by gender
- Average screen time by age group
- Addicted versus non-addicted distribution

Use the slicers first to define a population, then compare KPI cards and charts. The decomposition tree and Key Influencers visual are exploratory analytical tools rather than causal models.

7. Performance Testing

7.1 Utilization of Data Filters

Four distinct slicer/filter dimensions are visible: gender, stress level, addiction level and Age Group. These filters help users move from overall statistics to targeted segments.

7.2 No. of Calculation Columns / Measures

Two derived categorical fields are identifiable: Age Group and Addicted status. No separately named DAX measures are exposed in the report-definition queries inspected; the visuals primarily use direct Count, Average and Sum aggregations.

Item	Result
Derived/calculated fields visible	2 — Age Group; Addicted status
Explicit named measures visible in report queries	0 identified
Main aggregations	Count, Average and Sum

7.3 No. of Visuals

The PBIX contains multiple visual types across five pages, including KPI cards, bar/column charts, scatter charts, pie/donut charts, slicers, a decomposition tree, Key Influencers, line charts and text boxes.

8. Conclusion / Observation

The Smartphone Usage & Digital Well-Being project organizes a multi-variable smartphone dataset into an interactive Power BI analytics report. It covers usage time, entertainment behaviour, sleep, work/study, notifications, stress, demographics and addiction.

The combination of descriptive KPIs with interactive segmentation is a key strength. The scatter plot supports relationship exploration between screen time and sleep; demographic charts support group comparisons; and the decomposition tree and Key Influencers visual support deeper investigation.

The dashboard successfully delivers validated, high-precision metrics across all 7500 user profiles. The analysis establishes a clear baseline for digital habits: an average daily screen time of 7.50 hours escalates to a weekend average of 9.24 hours, with smartphone addiction affecting 5308 individuals (70.7% of the dataset). Advanced behavioral tracking captures an average of 97.83 app openings and 134.26 notifications per day, while predictive Key Influencers identify that a 2.61-hour increase in daily usage serves as the primary driver for high addiction risk. This comprehensive alignment between raw dataset aggregations, SQL validation queries, and interactive Power BI visual layers ensures full data integrity across the entire project repository.

9. Future Scope

- Add a formal data-quality page showing missing values, duplicates and outliers.
- Create explicit DAX measures for standardized KPIs and percentages.
- Add time-based analysis if a date/time field becomes available.
- Add correlation analysis and trend indicators.
- Add a mobile-friendly report layout.
- Add source and refresh-date documentation.
- Add a validated predictive model for addiction-risk exploration with appropriate ethical safeguards.

10. Appendix

10.1 Source Code (if any)

This project is implemented as a Power BI PBIX report. No separate external source-code file was identified inside the uploaded PBIX package.

10.2 GitHub & Project Demo Link

GitHub Repository: <https://github.com/durgadeviandrothu/Smartphone-Usage-Digital-Addiction-Well-Being-Analytics>

Project Demo / Power BI Link: <C:\Users\Durga\OneDrive\Desktop\smartphone.pbix>

Add the final public links before submission.

Appendix A – Dataset Fields Identified

Field	Role	Use in analysis
user_id	Identifier	User counting / KPI
transaction_id	Identifier	Record counting
age	Numeric	Age KPI / age grouping
gender	Categorical	Demographic comparison /

		slicer
daily_screen_time_hours	Numeric	Core usage KPI and comparisons
weekend_screen_time	Numeric	Weekend usage KPI
sleep_hours	Numeric	Sleep KPI / screen-time comparison
work_study_hours	Numeric	Productivity-related KPI
app_opens_per_day	Numeric	Usage intensity
gaming_hours	Numeric	Entertainment behaviour
social_media_hours	Numeric	Social-media behaviour
notifications_per_day	Numeric	Notification intensity
stress_level	Categorical	Stress comparison / slicer
addiction_level	Categorical	Addiction segmentation / slicer
addicted_label	Categorical/numeric label	Addiction status summary
academic_work_impact	Categorical	Decomposition analysis
Age Group	Derived category	Age segmentation
Addicted status	Derived category	Addicted/non-addicted segmentation

Appendix B – Submission Checklist

- Insert student/team details.
- Insert screenshots of Dashboard, Report 1 and Report 2.
- Refresh the PBIX and copy final KPI values if required.
- Add GitHub and project-demo links.
- Verify formatting, page numbers and spelling.